

VAIBHAV SETHI

Data Analyst

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PROFESSIONAL SUMMARY

Data analyst finishing a B.Sc. in Statistics and Physics at UBC, with two summers of paid analyst work at a market research firm cleaning, coding and analyzing client survey data in SPSS, SQL, Python and R. Builds projects end to end: the data pipeline, the model, then a dashboard or written report a non-technical team can use. Relocating to Toronto, available full time from January 2027.

CORE SKILLS

Programming and Databases: SQL (joins, CTEs, window functions), Python (pandas, NumPy, scikit-learn, XGBoost), R (tidyverse, ggplot2, Shiny), SAS, SPSS, DuckDB, Excel

Statistics and Modeling: Regression, logistic regression, generalized linear models, time series, clustering, PCA, survey analysis, cross-tabulation, machine learning, model validation (ROC, AUC), probability calibration

Reporting and Dashboards: Power BI, Tableau, Shiny, interactive web dashboards, data visualization, Excel reporting, written reporting for non-technical stakeholders

Tools and Workflow: Git, GitHub, Jupyter, RStudio, Parquet, ETL and data pipelines, data cleaning, feature engineering, documentation

PROFESSIONAL EXPERIENCE

Junior Data Analyst | BrookSphere Research

May 2024 - Sep 2024, May 2025 - Sep 2025

Market research firm running custom B2B and B2C studies, from survey design to reporting.

- Cleaned, coded and analyzed client survey data in SPSS across B2B and B2C studies for final client reports.
- Automated cleaning, tabulation and reporting steps in R and Python, cutting turnaround on recurring studies.
- Built cross-tabs, summary tables and charts that turned raw survey answers into findings clients could act on.
- Owned client data requests and revisions end to end against tight deadlines; asked back for a second summer.

Printing Services Representative | Staples Canada, Vancouver, BC

Dec 2022 - Jun 2023

- Completed 50 to 70 print orders per shift against fixed deadlines and resolved equipment failures on the spot.

SELECTED PROJECTS

Mortgage Portfolio Stress-Testing Engine | Python, SQL, DuckDB, R, SAS, XGBoost, Power BI

2026

Live dashboard and write-up: vaibhavsethi01.github.io/portfolio

- Estimated losses on a \$35B portfolio of 350,000 real US mortgages under Federal Reserve 2026 scenarios.
- Built the pipeline in SQL and DuckDB over 19 million loan-month records, from raw files to model-ready tables.
- Trained a default-prediction model, tested it on loan years it had never seen (Gini 0.66, KS 0.50), then calibrated it so scores read as real probabilities, cutting prediction error 63 percent.
- Projected losses over 13 quarters (\$65M baseline against \$352M severe) and shipped it as a live web dashboard with a plain English write-up, plus SAS and R versions and a Power BI export.

Credit Scoring and Expected Loss App | R, Shiny, XGBoost, SHAP

2024 - 2025

Code: github.com/vaibhavsethi01/banking-risk-platform

- Built a web app that scores loan applicants for default risk and explains in plain terms why each got their score.
- Benchmarked XGBoost against logistic regression on rare-event data using PR-AUC, calibrated the output, and added single and bulk CSV scoring for non-technical users.

Gravitational-Wave Noise Classifier | Python, scikit-learn, Random Forest

2026

- Sorted 7,966 detector noise samples into 22 categories on a 67-to-1 imbalanced dataset, reaching 88 percent accuracy and 0.81 macro-F1 after tuning with cross-validation, and confirmed results held on unseen data.

Vancouver Housing Price Prediction | R, Python, Time Series

2023

- Modeled home prices from 10 years of sales joined with rate, CPI and income data, validated by RMSE and MAE.

EDUCATION AND CERTIFICATIONS

University of British Columbia, Vancouver, BC

Coursework complete December 2026 | Convocation May 2027

B.Sc., Combined Major in Science: Statistics, Physics, Earth and Environmental Sciences

Relevant coursework: Regression and Generalized Linear Models, Probability, Statistical Inference, Machine Learning, Data Science in R and Python, Linear Algebra, Calculus

Certifications and activities: Bloomberg Market Concepts (BMC); UBC Data Science Club, member, 2025 to present